

Virtual Reality Lensing for Surface Approximation in Feature-Driven DVR

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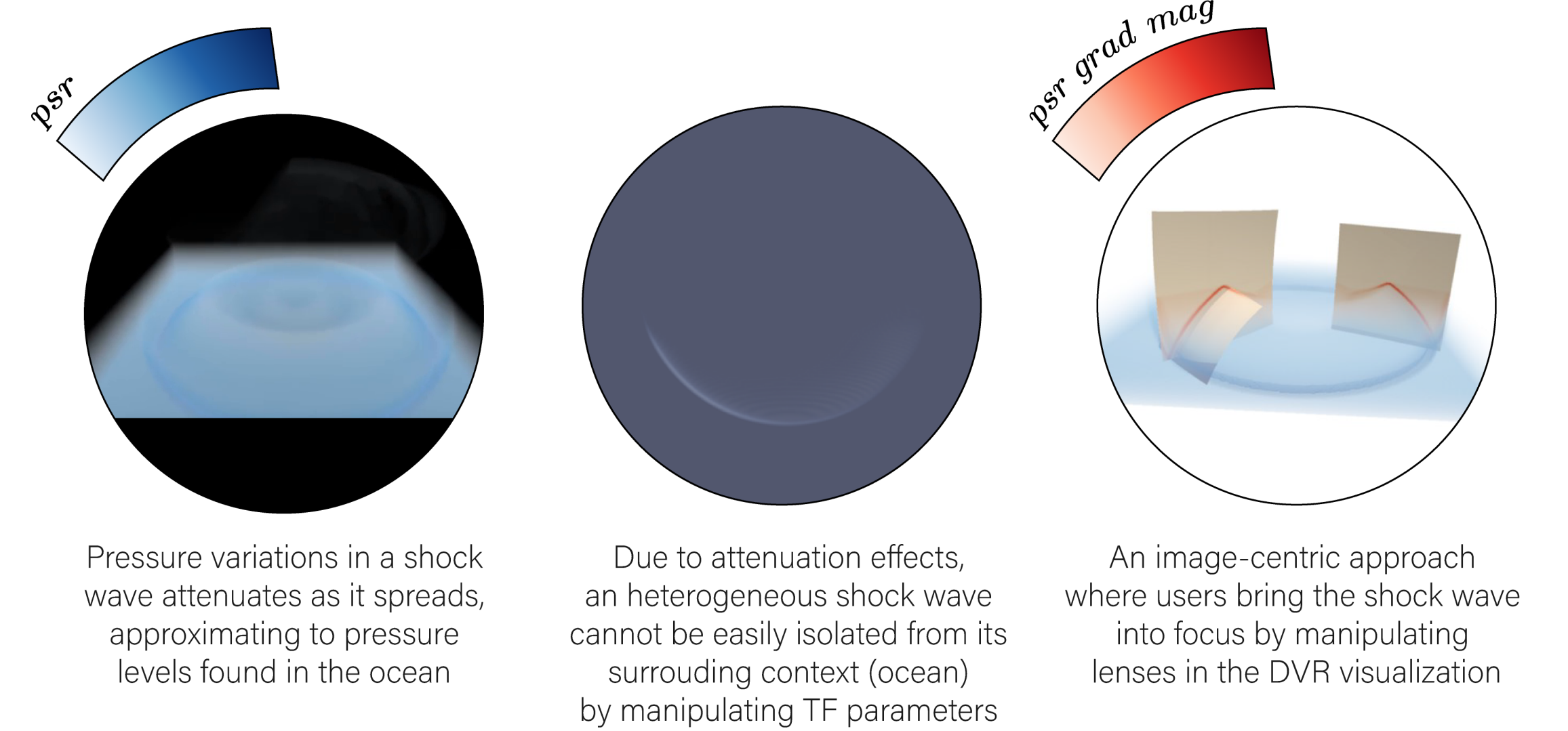
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INTRODUCTION

RQ How to bring visual focus to *heterogeneous* features in DVR visualizations?

1. **Data-driven**: specify feature in focus via indirect tuning of TF parameters
2. **Image-centric**: focus feature is set via direct interaction with the visualization



LENS DEFINITION & DESIGN

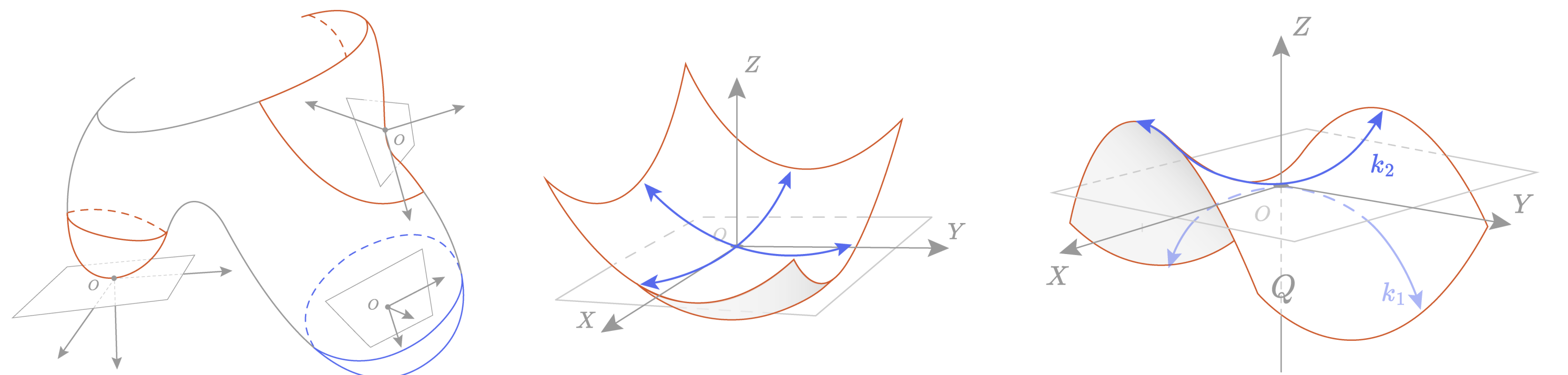
DEFINITION

The lens acts as a deformable **2D** manifold that emulates a *local* surface patch, based on the premise that a meaningful approximation of the feature can be built by composing multiple patches*.

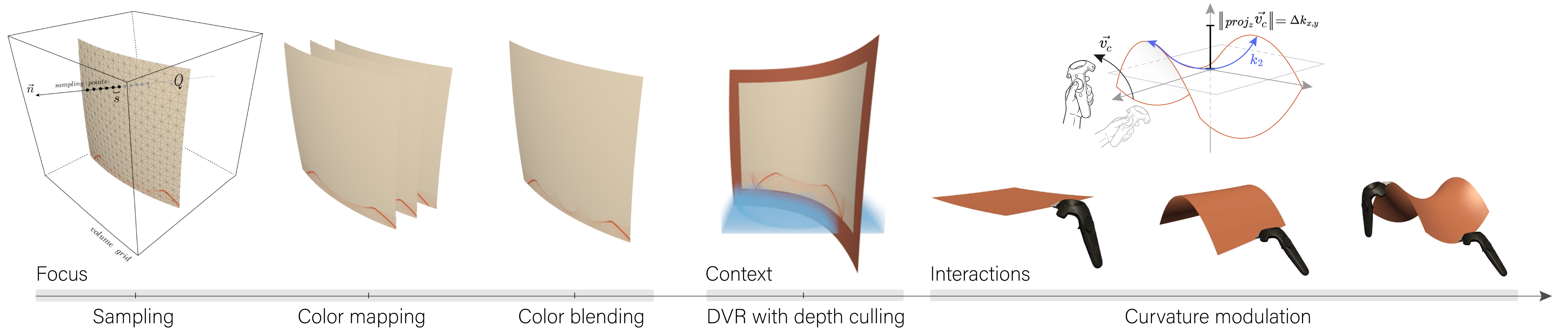
$$r(x, y) = [x, y, \frac{1}{2} (k_1 x^2 + k_2 y^2)]^T$$

The equation defines a quadric surface Q , where k_1 and k_2 coefficients are the two principal curvatures of the quadric—i.e. they measure the surface bending magnitude at each point $p \in Q$.

*do Carmo has proven that any smooth surface can be locally approximated by a differentiable function [2]



VISUAL & INTERACTION DESIGN



EXPERT FEEDBACK

- F₁** Multiple small > single large for approximation and context
- F₂** Role as alignment aids
- F₃** Locking against accidental edits
- F₄** Visual aids for control point disambiguation
- F₅** Standard DVR favored for spatial awareness

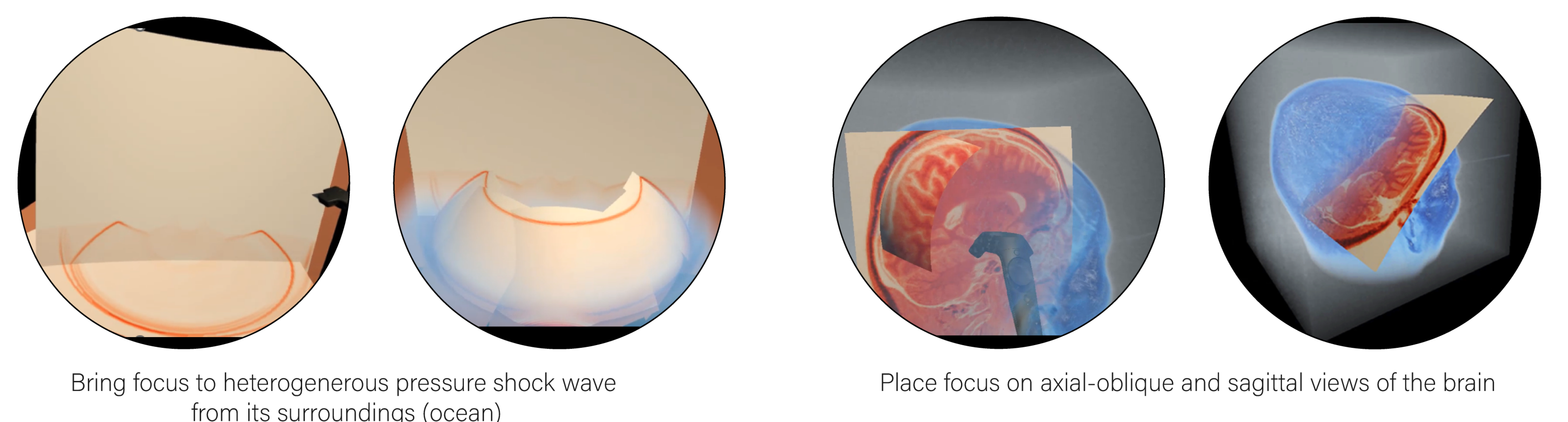
FUTURE WORK

1. Streamlined composition and editing of quadric images
2. Leverage descriptive spatial lines in reference guides

ACKNOWLEDGEMENTS

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PRELIMINARY ANALYSIS



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- [4] H. Pfister, B. Lorensen, C. Bajaj, G. Kindlmann, W. Schroeder, L. S. Avila, K. Martin, R. Machiraju, and J. Lee. The transfer function bakeoff. IEEE CG&A, 21(3):16–22, 2001.
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